

Background Mathematics: Finite Field Theory

Begin by looking at groups, rings, finite fields *used in cryptography:*

Definitions:

- Group *(can be finite or infinite)*
- Abelian Group
- Ring
- Commutative Ring
- Integral Domain
- Field

Group $\{G, \square\}$ $\square$ - binary operation

ex: $\{G, +\}$, $\{G, \times\}$
additive group, multiplicative group

1) Closure: $a \in G, b \in G \Leftrightarrow a \square b \in G$

2) Associativity: $a \square (b \square c) \Leftrightarrow (a \square b) \square c$

3) Identity element: $\exists e \in G$ s.t. $a \square e : e \square a = a \forall a \in G$

4) Inverse element: $\forall a \in G, \exists a' \in G$ s.t. $a \square a' = a' \square a = e$

"order" of a group: # elements in G .

First 4 properties form a group.

5) Commutativity: $a \square b = b \square a \forall a, b \in G$

First 5 properties form an Abelian group.

"exponentiation": repeated application of the group operator

e.g. $a^3 = a + a + a$
(in an additive group)

e.g. repeated addition

^{whatever the operator happens to be.}

G is cyclic if every element of G is a power a^k of a fixed element " a " $\Rightarrow$ " a " is a generator of G .
 $\hookrightarrow$ happens to be a power of a particular element ' a '.

Ex: Integers modulo 5: 0, 1, 2, 3, 4

Note: 0 is not a power of any element, so 0 cannot be part of a cyclic group. $\{1, 2, 3, 4\}$ are part of a cyclic group.

Ring: $\{R, +, \times\}$ (2 binary operators)
Set of elements 2 operators
(typical in crypto add & mult)
 - satisfies the above 5 properties for addition operations.

6) Closure under multiplication

7) Associativity under multiplication.

8) Distributivity: $a(b+c) = ab+ac, \forall a, b, c \in R$

9) If R also satisfies commutativity for multiplication, then it is a generator.
 called a commutative ring.

(usually 1)

10) If there is a multiplicative identity $a \cdot 1 = 1 \cdot a = a \forall a$ and no "zero divisors" (if $a \neq 0, a, b \in R$ and $a \cdot b = 0$ then $a = 0$ or $b = 0$)
 $\Rightarrow$ Then R is called an integral domain.

What is a zero divisor

Field: $\{F, +, \times\}$

- satisfies the above 10 + (it is an integral domain)

11) multiplicative inverse $\forall a \in F \exists a' \in F$ s.t. $a'a = a \cdot a' = 1 =$ mult. identity.

$\uparrow$ For every element a in the Field, there is another element a' in the Field also.
 s.t. $a \cdot a' = 1 =$ gives us the multiplicative identity.

Fig 4.2

Typically in crypto, we are going to look at $+$ & $\times$, but theoretically any 2 operators satisfies the first 5 under the first opr and then the others under the other operator.

Ex

$$\begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix} \bmod 5 = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix} \quad \begin{pmatrix} 3^0 \\ 3^1 \\ 3^2 \\ 3^3 \\ 3^4 \end{pmatrix} \bmod 5 = \begin{pmatrix} 1 \\ 3 \\ 4 \\ 2 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} 2^0 \\ 2^1 \\ 2^2 \\ 2^3 \\ 2^4 \end{pmatrix} \bmod 5 = \begin{pmatrix} 1 \\ 2 \\ 4 \\ 3 \\ 1 \end{pmatrix} \quad \begin{pmatrix} 4^0 \\ 4^1 \\ 4^2 \\ 4^3 \\ 4^4 \end{pmatrix} \bmod 5 = \begin{pmatrix} 1 \\ 4 \\ 1 \\ 4 \\ 1 \end{pmatrix}$$

$\{2, 3\}$ gens all non-zero elements in the group

$\Rightarrow$ that means 2 & 3 are generators of G , and 1 & 4 are not generators of the group.
 $\Rightarrow$ so, the non-zero elements in G form a cyclic group and the element(s) whose powers generate all of them, that element is known as a generator.
 $\therefore$ 2 & 3 are both generators of this group.
 $\therefore$ 1 & 4 are not generators

fig 4.2

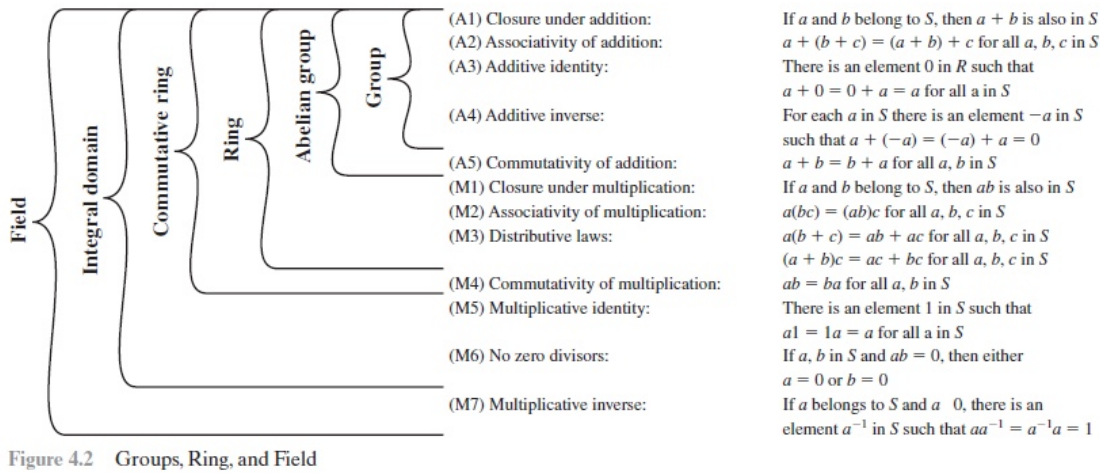


Figure 4.2 Groups, Ring, and Field

(see Stallings, Figure 4.1)

Modular arithmetic

- $a \bmod n$ The remainder ("residue") when a is divided by n
residue class: Set of remainders as we take all the elements in the group or elements in the field and divide by N .
- congruency a is congruent to $b \bmod n$ ($a \equiv b \bmod n$)
if $a \bmod n = b \bmod n$ $6 \bmod 5 = 11 \bmod 5 (=1)$
low: (their residues are equal)
- b divides a if $a = mb$ for some m , where a, b, m are integers
written as $b|a$

huge implementation concept → • notation: use of mod operator
 $(a+b) \bmod n = [(a \bmod n) + (b \bmod n)] \bmod n$

- Z_n : the set of non negative integers less than n .
↳ that just means integers modulo n

$$Z_n = \{0, 1, 2, \dots, n-1\} \text{ "set of residues"}$$

→ We can always do addition, subtraction & multiplication in Z_n
→ But division is not always possible because some elements do not have a multiplicative inverse
(see Stallings, Table 4.1)

(additive inverse)

Greatest Common Divisor, gcd

- $\gcd(a, b) = \max [k, \text{such that } k|a \text{ and } k|b]$
 a, b are "relatively prime" is $\gcd(a, b) = 1$
 a & b don't have to be prime

Euclidean algorithm for finding gcd: for positive integers a, b
where $a > b$

- $\gcd(a, b) = \gcd(b, a \bmod b)$

Note: all non zero int. divide 0

$$\Rightarrow g(a, 0) = a.$$

$$\gcd(18, 12) = \gcd(12, 6)$$

$$\gcd(12, 6) = \gcd(6, 0) = 6$$

► Finite field $\mathbb{Z}_n$ is usually an integral domain.
 ← integers modulo something

- In general, $\mathbb{Z}_n$ is not a field (not necessarily a field) because not every element has a multiplicative inverse.

$\mathbb{Z}_p$: every element is relatively prime to p .
 $\Rightarrow$ every element has a multiplicative inverse.

$\therefore \mathbb{Z}_p$ is a field otherwise it would just be a ring/semi-field
 (always)
 $\mathbb{Z}_n$ is not a field if n happens not to be prime

► it turns out that the only elements that have a multiplicative inverse, are elements that are relatively prime to the modulus. (an element that has no factors in common with the modulus other than 1).

- It can be shown that the order of a field must be a power of a prime

p^n for some positive int. n $\hookrightarrow$ number of elements of a field

The integers modulo P will definitely form a field $\mathbb{Z}_P$ prime

($\Rightarrow$)

every element in that set is relatively prime to modulus

◦ Galois Field: $\text{GF}(p^n)$

$\hookrightarrow$ Finite Field $\mathbb{Z}_p$

the only elements that are relatively prime to the modulus have a multiplicative inverse

$\Leftrightarrow$
 in other words

$$\text{GCD}(a, N) = 1$$

what relatively prime means they have no factors in common they are prime with each other.

$\mathbb{Z}_p$
 $\downarrow$

Two special cases: $n=1$, and $p=2$ [$\text{GF}(p)$ and $\text{GF}(2^n)$]

$$\text{GF}(p) = \mathbb{Z}_p \{0, 1, 2, \dots, p-1\}$$

► How many elements in a field / (order of a field)
 order of a field will always be the power of a prime.

in a field every element has a multiplicative inverse

The Euclidean algorithm for finding $\gcd(a, b)$ can be extended to find the multiplicative inverse of $\underline{b}$ modulo $\underline{a}$, if it exists

Ext^d Euclidean algo, $\gcd(a, b)$

Find s, t such that $\underline{sa} + \underline{tb} = \gcd(\underline{a}, \underline{b})$

Example: $\gcd(161, 28)$

by convention, a is larger num

$$s_3 = s_1 - 2s_2$$

$$t_3 = t_1 - 2t_2$$

$$161 = 5(28) + 21$$

When $r = 0$ $\gcd(a, b) = b$ value on the last row
(the row that has the 0 remainder in it.)

$s = s_2$ on last row

$t = t_2$ on last row

$\gcd(161, 28)$

Euclidean part

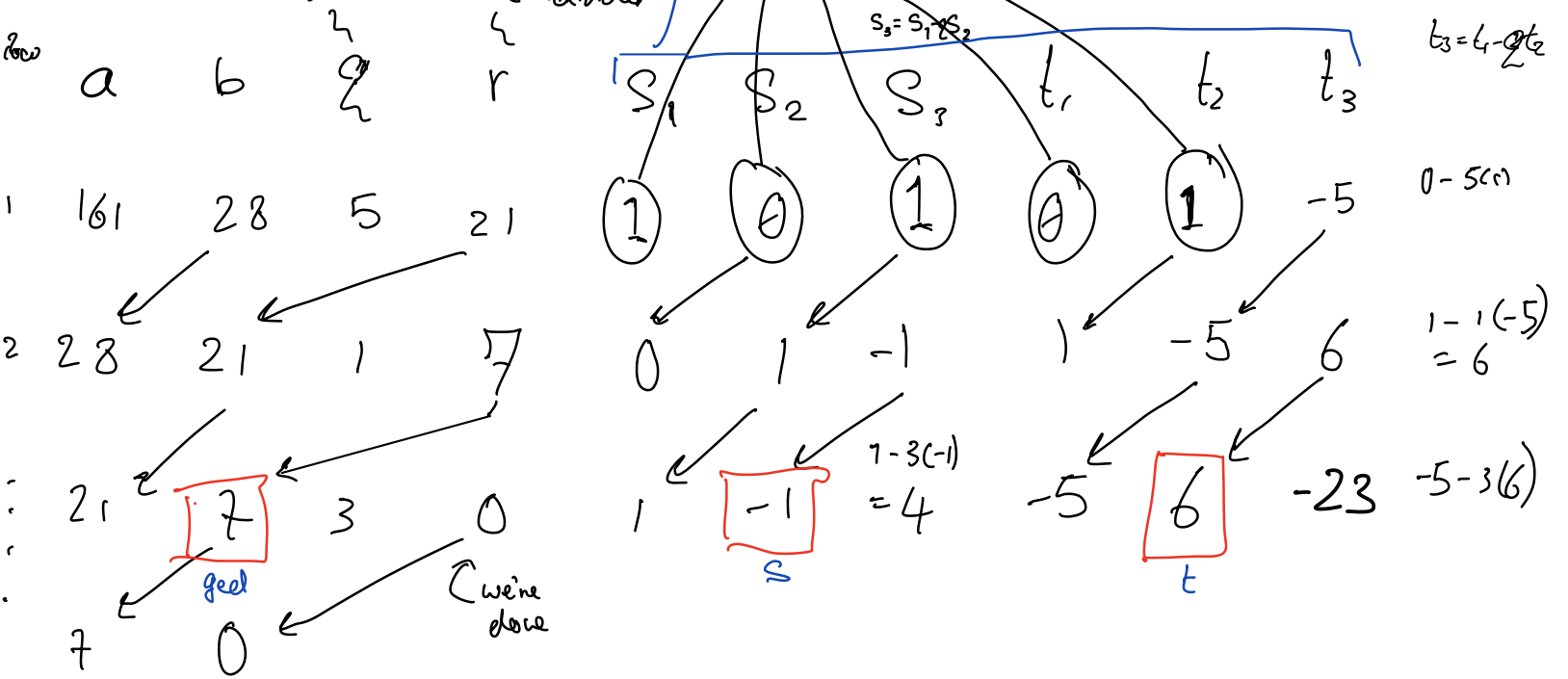
Quotient

remainder

don't need s and t 's if doing
just the Euclidean (not the

Extended-Euclidean)

Starting variables (when filling out first row)
 s_3 is going to be 1

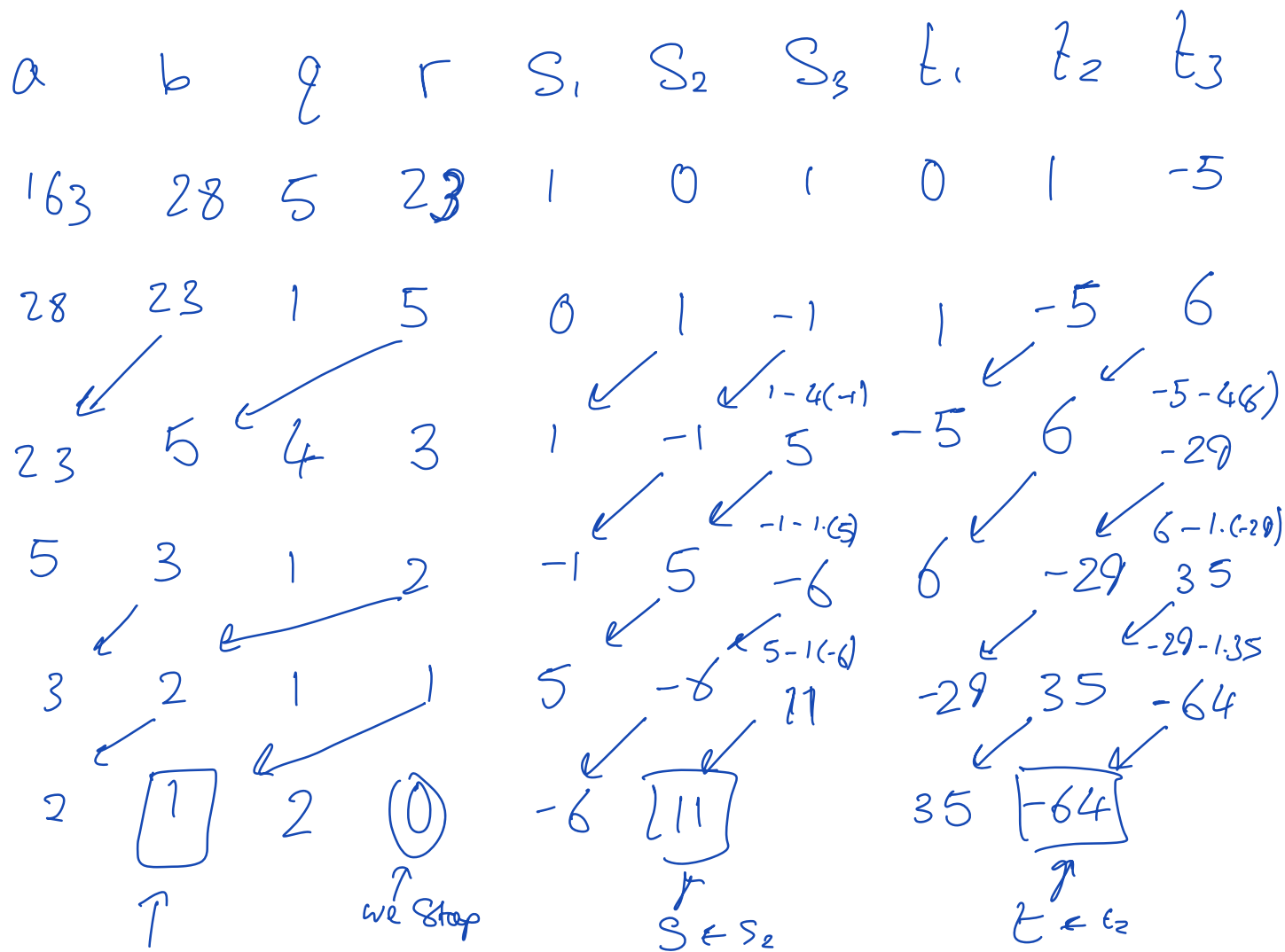


$$sa + tb = \gcd(a, b)$$

$$-1(161) + 6(28) = \underline{7}$$

~~our~~

Ex #2. $\gcd(163, 28)$ ^{Prime number, so we know its going to be a gcd of 1.}



$$S a + t b = \gcd(a, b)$$

$$11(163) + (-64)(28) = 1$$

Our S value is 11

Our t value is the multiplicative inverse

$$t = -64$$

$$t \bmod 163 = 99$$

$$-64 \bmod 163 = 99$$

$$(99 \times 28) \bmod 163 = 1$$

So 99 is the multiplicative inverse of $6 \bmod a$.
 $(28 \bmod 163)$

Euclidean Algorithm works not only with numbers but also with Polynomials!
 we can take 2 Polynomials and find their gcd, which will be a Polynomial.
 if they have no factors in common, their result will be 1.
 we can find the multiplicative inverse of a Polynomial modulo another Polynomial.

Polynomial Arithmetic

3-levels

1. Ordinary polynomial arithmetic
2. Polynomial arithmetic in which the arithmetic on the coefficients is performed modulo p (i.e., in $\text{GF}(p)$)
3. Polynomial arithmetic with coefficients in $\text{GF}(p)$ and polynomials are defined modulo a polynomial whose highest power is n

1. Ordinary poly arith.: example

Secondary School arithmetic.

$$F(x) = x^3 + x^2 + 2$$

$$G(x) = x^2 - x + 1$$

$$F(x) + G(x) = x^3 + 2x^2 - x + 3$$

$$F(x) - G(x) = x^3 + x^2 + x + 1$$

$$F(x) \cdot G(x) = x^5 + 3x^2 - 2x + 2$$

$$C_K \neq 3$$

Ex 3

↑
geel

$$E_i =$$

$$Q = u_A,$$

Example 4. find $\gcd(x^3+x+1, x^2+x)$
also mult. inv of b mod a
 (x^2+x) mod x^3+x+1

	a	b	c	r	s ₁	s ₂	s ₃	t ₁	t ₂	t ₃
	$x^3 + x + 1$	$x^3 + x$	$(x+1)$	1	1	0	1	0	1	$x+1$
	$x^2 + x$	$\boxed{x}$	$x^2 + x$	0	0	$\downarrow \begin{matrix} \text{II} \\ \text{I} \end{matrix}$		$\downarrow$	$\boxed{x+1}$	
		good				S				
		wrong								
		leading term = 1								

want
 leading term = 1
 coefficient by dot

multiverse

t_{mod}'

$$= (2x+1) \bmod (x^3+x+1)$$

$$(x+y) \times (x^2+yx) = x^3+yx^2$$

$$n \equiv r \pmod{a}$$

$$\Rightarrow x^3 + x \bmod (x^3 + x + 1) = 1$$

$$\begin{array}{r}
 x+1 \leftarrow g_0 \\
 \hline
 \left. \begin{array}{l} x^2 + x + 1 \\ x^3 + x^2 \end{array} \right\} x^2 + x + 1 \\
 \hline
 x^2 + x + 1 \\
 \hline
 x^2 + x \\
 \hline
 x \leftarrow g_1
 \end{array}$$

$$\begin{aligned}
 t_3 &= 0 - (a_{x+1}) \\
 &= -n g_{x+1} \\
 &= -x
 \end{aligned}$$

2. Poly arith with coeff. in $GF(p)$: example

> any operations are done on the coefficients and performed modulo p . polynomial ring (not a field).
 division is not always possible

- (see Stallings, Figure 4.4 for example over $GF(2)$)

e.g.:

$$x^7 + x^5 + x^4 + x^3 + x + 1$$

$$x^3 + x + 1$$

XOR 2 Polynomials

$$x^7 + x^5 + x^4 + (x^3 + x + 1) \text{ mod } (x^3 + x + 1)$$

- Can use Euclidean alg. over the polynomials

addition & subtraction are the same in $GF(2)$
 XOR-Operation

See Fig 4.4

- Can find something analogous to a prime number
 - Prime polynomial (irreducible polynomial)

Ex: $x^4 + 1$ over $GF(2)$ is ^{not irreducible} reducible because

$$x^4 + 1 = (x + 1)(x^3 + x^2 + x + 1)$$

$$x^4 + 1 = (x^2 + 1)(x^2 + 1)$$

Ex: $x^3 + x^2 + 1$ over $GF(2)$ is irreducible (i.e. a prime polynomial)

$x^4 + 1$ is a composite polynomial, not a prime polynomial.

3. Poly arith (with coeff. in $\text{GF}(p)$), modulo a polynomial ^{irreducible}

Forms a finite field

Recall: $\mathbb{Z}_p$ is a field

Why don't we just use $\mathbb{Z}_p$? :
- complete encoding: represent all values of interest.

Motivation for this is implementation

- complete encoding (can represent all values of interest) and efficient processing

e.g. We are encrypting data 8 bits at a time
→ can't do everything in integers $\mathbb{Z}_{256}$ because this is not a field.
→ The largest prime less than 256 is 251
→ $\mathbb{Z}_{251}$ (can't use ^{bit} patterns that represent 251-255).

→ We want a field with exactly 256 elements.

Turn set of polynomials into a field

Consider the set of polynomials of the form
 $f(x) = a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + \dots + a_1x + a_0 = \sum_{i=0}^{n-1} a_i x^i$

If each a_i takes a value in $\{0, \dots, p-1\}$, then there are p^n such polynomials. But we know that a field has p^n elements.

Define: arithmetic follows ordinary rules of polynomial arith with 2 refinements:

- 1) Arith. on coefficients is done mod p
- 2) If multiplication results in a poly. $f(x)$ of degree greater than $n-1$, then reduce mod. an irreducible polynomial $m(x)$
 $r(x) = f(x) \bmod m(x)$.

Poly

(see Stallings, Table 4.6 for example over $GF(2^3)$)

Poly. arith. over $GF(2)$ mod $x^3 + x + 1$

$$(101) \cdot (111) \\ x^2 + 1 \cdot x^2 + x + 1$$

$$\begin{array}{r} (x^2 + x + 1) \\ (x^2 + 1) \times \\ \hline x^2 + x + 1 \end{array}$$

$$\begin{array}{r} x^4 + x^3 + x^2 \\ \hline x^4 + x^3 \\ \hline \end{array}$$

$$\begin{array}{r} x^3 + x + 1 \quad \begin{array}{r} x + 1 \\ \hline x^4 + x^3 + x + 1 \\ - x^4 + x^2 + x \\ \hline x^3 + x^2 + 1 \\ x^3 + x + 1 \\ \hline x^2 + x \end{array} \\ \hline \end{array}$$

Remainder.
 $x^2 + x$
 (110)

$$101 \times 111 = 110 \text{ mod } 1011$$

Note this is not

$$5 \times 7 = 6 \text{ mod } 11$$

→ We are treating the bit string as polynomials (not integers)

Can use extended Euclidean alg. over polynomials to find multiplicative inverse of a polynomial

So, we can have complete encoding. What about efficient processing?

XOR: $\begin{matrix} 0 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{matrix}$

Choose $p=2$.

- Addition and subtraction is simply XOR

- Multiplication is left-shift and (conditional) XOR

$$f(x) = a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + \dots + a_1x + a_0$$

$$\therefore f(x) \text{ times } x = a_{n-1}x^n + a_{n-2}x^{n-1} + \dots + a_1x^2 + a_0x + 0$$

This is a left shift of the bit string by 1 bit.

If $a_{n-1} = 1$, then this is a polynomial of degree n , so it must be reduced modulo $m(x)$.

$$m(x) = x^n + m_{n-1}x^{n-1} + \dots + m_1x + m_0$$

but $x^n \bmod m(x)$ is equal to $m(x) - x^n$

$$\begin{array}{r} \overline{m(x)} \overline{x^n} \\ - m(x) \\ \hline m(x) - x^n \end{array}$$

$$= m_{n-1}x^{n-1} + \dots + m_1x + m_0$$

$\therefore$ If $a_{n-1} = 1$, then substitute $(m_{n-1}x^{n-1} + \dots + m_1x + m_0)$ for x^n

$$f(x) \text{ times } x = \begin{matrix} a_{n-2} & a_{n-3} & \dots & a_1 & a_0 & 0 \\ \text{if } a_{n-1} = 0 & & & & & \\ a_{n-2} & a_{n-3} & \dots & a_1 & a_0 & 0 \oplus \\ \leftarrow m_{n-1} & m_{n-2} & \dots & m_1 & m_0 & \\ \text{if } a_{n-1} = 1 & & & & & \end{matrix}$$

except leading coeff. of irr. poly

- Division is multiplication with the inverse

$f(x) \div g(x)$ use euclidean alg. to find $g^{-1}(x)$
 $f(x)$ times $g^{-1}(x)$

$$a^b \bmod n$$

Recall

$$\mathbb{Z}_5 = \{0, 1, 2, 3, 4\}$$

-generators of the cycle of non-zero elements are 2 & 3

$$\left(\begin{array}{l} \text{recall: } 2^0 = 1, 2^1 = 2, 2^2 = 4, 2^3 = 8 = 3, 2^4 = 2^0 = 1, \dots \\ 3^0 = 1, 3^1 = 3, 3^2 = 9 = 4, 3^3 = 27 = 2, 3^4 = 3^0 = 1, \dots \end{array} \right)$$

$$x \pmod{5}$$

	0	1	2	3	4
0	0	0	0	0	0
1	0	1	2	3	4
2	0	2	4	1	3
3	0	3	1	4	2
4	0	4	3	2	1

$\Rightarrow$

$$x \pmod{5}$$

	0	1	2^1	2^2	2^3	2^4
0	0	0	0	0	0	0
1	0	1	2_2	2^3	2^4	2^5
2	0	2	2	1	2^3	2^4
2^3	0	2^3	1	2^2	2	2^3
2^2	0	2	2^2	2^3	2	1

$$\Rightarrow$$

	0	1	3^3	3	3^2
0					
1					
3^3					
3					
3^2					

$(\pmod{5})$

In general, generator g

	0	1	g	g^2	g^3
0					
1					
g					
g^2					
g^3					